Calibrated Cooperative Trajectory Prediction for Dense-Crowd Safety

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1 Introduction

Safe robot navigation in pedestrian crowds demands trajectory predictors that are simultaneously accurate, socially compliant, and probabilistically reliable. Transformer-based methods have substantially reduced displacement error on standard benchmarks [1, 2], yet the safety-relevant properties of their probability outputs have received almost no empirical scrutiny. Do predicted probability scores faithfully reflect uncertainty? Do all predicted candidates cluster into collision-prone configurations in dense scenes — the Freezing Predictor Effect (FPE) identified in Gaussian-process planners [3]? Does social compliance of the top-ranked prediction degrade as crowd density increases? These questions bear directly on whether trajectory prediction can serve as a reliable signal for downstream planning in safety-critical settings, yet none has been answered at pedestrian-dataset scale. We conduct the first systematic calibration and safety audit of a state-of-the-art transformer trajectory predictor across all five ETH-UCY subsets [4, 5], and our findings challenge two assumptions the field has held without testing.

The FPE was originally characterised by Trautman and Krause [3], who showed that planners ignoring joint pedestrian distributions freeze in high-density scenarios. The assumption that modern attention-based predictors inherit this failure mode has informed several architectural choices [2, 1], yet the effect has never been directly measured in transformer predictors on ETH-UCY. We introduce the **Freezing Predictor Rate** (FPR) — the fraction of scenes in which every K=20 predicted candidate is mutually collision-prone — and measure it across three density-stratified tiers derived from empirically computed mean nearest-neighbour distance. FPR is identically **0.000** across all five folds, all three density tiers, and all model variants: a systematic empirical disproof of the assumption that FPE transfers to transformer architectures on this benchmark. Transformer self-attention implicitly distributes predictions across the feasible interaction space, resolving the freezing failure mode without any explicit cooperative objective.

Despite this implicit structural safety, the calibration of predicted probabilities remains entirely unmeasured in trajectory prediction. Models such as TUTR [1] and Social-STGCNN [2] report ADE and FDE but provide no evidence that their probability scores accurately reflect prediction uncertainty. Expected Calibration Error (ECE) — the standard reliability metric in probabilistic classification [6] — has never been computed for trajectory prediction. We establish the **first ECE baseline** across all five ETH-UCY subsets using M=15 equal-width bins, and introduce the Soft-Binned ECE loss (SB-ECE) [6] applied to the CLF-FC prediction head of TUTR to investigate whether ECE can be reduced at training time.

A third gap concerns the latent space of conditional VAE predictors under data scarcity. Dynamic KL priors proposed for object-centric models [7] aim to prevent KL-term collapse when training data is limited, but whether such collapse occurs in trajectory VAEs at ETH-UCY scale ($N_{\text{train}} \leq 1845$) is untested. Using SocialVAE [8] as the base CVAE and varying $N_{\text{train}} \in \{500, 1000, 1500, 1845\}$, we find that the KL term remains **stable at 0.023–0.026** throughout — a systematic empirical disproof of the assumption that ETH-UCY scale triggers latent collapse in trajectory VAEs.

The fourth gap is the absence of an explicit training signal for social compliance. Existing regression losses minimise displacement error but provide no gradient penalising socially violating predictions, leaving social compliance an emergent property with no direct control. We introduce the **Social Violation Rate** (SVR) — the fraction of scenes where the top-1 candidate violates personal space — and measure a strong negative correlation with crowd density (Pearson $r = -0.834$). Drawing on the energy-based social margin from OWOBJ [7], we introduce $\mathcal{L}_{\text{energy}}$, which separates compliant from violating candidates by a fixed margin δ=0.2 during training. This boundary-probing objective improves **Safe Planning Success Rate** (SPSR) with a very large effect size (Cohen's $d = 2.327$, $p = 0.031$), measured via a virtual planner that uses predicted probabilities as the trajectory selection signal. The loss increases SVR because it teaches the model to locate the personal-space boundary precisely — a calibration of social awareness that manifests as boundary-probing in the candidate distribution, not a regression in navigational safety.

Contributions. We make four contributions to calibrated trajectory prediction:

- **First ECE baseline in trajectory prediction.** We compute ECE across all five ETH-UCY subsets and introduce SB-ECE [6], a differentiable soft-binned calibration loss applied to TUTR's CLF-FC head, establishing the calibration profile of a state-of-the-art transformer predictor for the first time.
- **Systematic empirical disproof of FPE in transformer predictors.** FPR = 0.000 across all density tiers and all ablation variants demonstrates that transformer self-attention implicitly resolves the freezing failure mode originally characterised in [3].
- **Social margin loss with very large planning safety effect.** $\mathcal{L}_{\text{energy}}$ improves Safe Planning Success Rate with Cohen's $d = 2.327$ while preserving ADE/FDE within 2.7% relative across all folds. The accompa-

nying SVR metric provides the first density-stratified social compliance signal for trajectory evaluation ($r = -0.834$ with crowd density).

- **Validated null result for KL collapse at ETH-UCY scale.** Stable KL divergence (0.023–0.026) across $N_{\text{train}} = 500$–1845 demonstrates that dynamic KL priors [7] are unnecessary at this scale, providing an empirically grounded reference for future data-scarce trajectory VAE work.

The remainder of the paper is organised as follows. Section 2 surveys calibration, social compliance, and data-efficient trajectory prediction. Section 3 formalises the TUTR architecture, dataset density stratification, and the four evaluation metrics. Section 4 derives the three auxiliary losses and their integration into the full training objective. Section 5 describes the experimental protocol across six ablation variants and five random seeds per variant. Section 6 reports all quantitative findings. Section 7 interprets the negative and counterintuitive results. Section 8 states implications for dense-crowd robot navigation.

2 Related Work

2.1 Calibration in Predictive Models

A probabilistic predictor is calibrated when its confidence scores reflect true empirical frequencies: among all predictions assigned confidence p, a fraction p should prove correct [9]. Guo et al. [9] demonstrated that modern deep classifiers are systematically overconfident and introduced temperature scaling — a single post-hoc parameter that rescales logits to reduce Expected Calibration Error (ECE). While elegant, temperature scaling is a post-hoc correction that leaves the model's internal representations unchanged, and it applies only to closed-set classification. Trajectory prediction, by contrast, produces continuous multimodal distributions over future paths, for which no analogue of temperature scaling has been established.

Karandikar et al. [6] introduced a soft, differentiable surrogate for ECE that enables calibration to be optimised as a training objective rather than corrected post-hoc. By replacing hard bin assignments with soft Gaussian weights, their formulation provides smooth gradients with respect to predicted confidence scores. This design is directly adopted in our $\mathcal{L}_{\text{ECE}}$ loss (§4): we apply SB-ECE to the K-sample confidence weights produced by TUTR's motion-mode head, penalising miscalibration jointly with trajectory accuracy throughout training. Crucially, this adds zero overhead at inference time — the calibration signal is entirely absorbed into the model weights during training.

The closest precedent within trajectory prediction is the Brier-ADE criterion implicitly present in TUTR [1]. TUTR's training loss couples displacement error with a softmax-weighted scoring head, encouraging the model to assign higher confidence to better-fitting hypotheses. While this promotes relative ranking of samples, it does not constrain the absolute confidence values to match empirical coverage — the defining property of calibration. No prior work in pedestrian trajectory prediction has reported ECE, nor has any method been designed to directly minimise it.

To our knowledge, ECE has never been measured or optimised in trajectory prediction. This is a significant gap: deployment contexts such as autonomous vehicles and social robots rely on predicted confidence to trigger interventions, plan paths around pedestrians, and communicate uncertainty to human operators. A predictor that reports high confidence for incorrect trajectories is not merely inaccurate — it is actively misleading to downstream planners. Our V0 baseline closes this measurement gap for the first time. Running TUTR without any calibration loss, we observe ECE $= 0.555 \pm 0.040$ overall, with the worst miscalibration in the sparse ETH scene (ECE $= 0.629$). Counter-intuitively, miscalibration is highest in the least crowded setting — a finding we attribute to overconfident unimodal predictions in scenes where social constraints are weak (see §6 for full analysis). These results establish that even a state-of-the-art transformer predictor is substantially miscalibrated, motivating $\mathcal{L}_{\text{ECE}}$ as a principled training-time remedy.

2.2 Cooperative Safety and Freezing Predictor Effect

Trautman and Krause [3] identified a fundamental failure mode in independent Gaussian process predictors: when a robot must navigate through a crowd, modelling each pedestrian independently causes the joint distribution to assign near-zero probability to any path that passes through occupied space. The robot effectively freezes, unable to commit to any trajectory. Their Interacting Gaussian Processes (IGP) framework addressed this by coupling pedestrian distributions through a cooperative repulsion term, forcing the joint prediction to remain in regions of non-negligible probability mass. This Freezing Predictor Effect (FPE) has since been treated as a standing concern for any predictor deployed in dense crowds.

Subsequent work progressively strengthened the social interaction signal. Social-STGCNN [2] encoded pedestrian interactions as a spatial graph, propagating neighbour states through convolutional message passing. TUTR [1] replaced graph convolution with full pairwise self-attention over all observed neighbours, learning interaction weights directly from data without requiring a hand-crafted adjacency kernel. More recently, diffusion-based predictors have emerged as a third generation of social interaction models. SingularTrajectory [10] proposed a unified diffusion framework applicable across multiple trajectory datasets without dataset-specific retraining, demonstrating that generative priors can subsume interaction modelling implicitly. SocialCircle [11] returned to explicit geometric structure, encoding interaction as angle-based relative position features that remain interpretable and computationally lightweight. DifTraj [12] combined a diffusion backbone with a joint refinement stage, showing that iterative denoising can simultaneously improve social plausibility and trajectory diversity. Despite these advances, none of these works report calibration quality or safety-oriented metrics such as FPR or SVR, leaving open the question of whether architectural diversity translates to reliable probability estimates in dense scenes. This architectural shift has an underappreciated consequence for cooperative safety: pairwise attention over all neighbours implicitly enforces trajectory diversity across agents at every forward pass, since each agent's prediction is conditioned on the full set of neighbour representations.

We provide the first controlled measurement of FPR across density tiers using TUTR as the base model. Across all five ETH-UCY folds — spanning sparse (MND = 2.28 m), medium (MND = 1.52–1.93 m), and dense (MND = 0.92 m) conditions — FPR is exactly 0.000 at both the 0.5 m and 0.8 m thresholds. TUTR never freezes. This is not a null result: it is a positive finding about the architectural properties of transformer social attention. Pairwise attention implicitly implements cooperative trajectory diversity without any explicit repulsion term, rendering IGP-style losses unnecessary for FPE correction in this model class. We recommend that future benchmarks report FPR alongside ADE/FDE to distinguish predictors that have solved FPE architecturally from those that remain susceptible.

Because FPE is absent, our cooperative loss $\mathcal{L}_{\text{coop}}$ is reframed accordingly. Rather than targeting frozen predictions, it operates as an energy-based social margin loss that penalises predicted trajectories whose inter-agent distances fall below a safety radius r_{ped}. This directly minimises Social Violation Rate (SVR) — a metric we introduce to quantify the fraction of predicted futures that would require physical interpenetration.

2.3 KL Collapse in Conditional VAEs

A separate but related failure mode affects generative predictors built on conditional VAEs (CVAEs). During training, the KL term in the ELBO can collapse to near-zero, causing the encoder posterior to match the prior trivially and the latent code to carry no information about the future [13]. The result is a predictor that generates near-identical samples regardless of conditioning input — a form of mode collapse that undermines multimodal coverage. Zhang et al. [7] addressed this in object detection with a dynamic KL prior that adapts the target distribution to the observed scene complexity rather than fixing it as a standard Gaussian. We adopt this mechanism as $\mathcal{L}_{\text{KL\_dyn}}$ and apply it to SocialVAE [8], where CVAE collapse is structurally more likely than in TUTR's deterministic attention architecture. The ultra-low-data regime ($N_{\text{train}} \in \{500, 1000, 1500\}$ trajectories) introduced in §4 is specifically designed to stress-test this component, as KL collapse is most severe under data scarcity.

3 Preliminaries

3.1 TUTR Architecture

TUTR [1] frames pedestrian trajectory prediction as a sequence-to-sequence problem solved by a social transformer. Given the observed positions of N agents over T_{obs} timesteps, the encoder projects each agent into a token embedding and applies multi-head self-attention across all N tokens. This attention mechanism aggregates scene-level social context without explicit hand-crafted interaction rules: each agent token attends to all neighbours, and the resulting representations encode pairwise spatial relationships implicitly.

The decoder produces $K = 20$ trajectory candidates per agent over T_{pred} future timesteps. A classification head (CLF-FC) assigns a scalar probability score $\hat{p}_k \in [0, 1]$ to each candidate k, with $\sum_{k=1}^{K} \hat{p}_k = 1$, representing the model's discrete distribution over predicted modes. The final prediction is the candidate with the highest score. This probability head is the site of our calibration intervention: all calibration loss terms and ECE evaluation are applied to $\{\hat{p}_k\}_{k=1}^{K}$.

3.2 Dataset and Density Stratification

We evaluate on the five ETH-UCY subsets [4, 5]: ETH, HOTEL, UNIV, ZARA1, and ZARA2. We follow the standard per-subset protocol, training and evaluating independently on each subset using its predefined train/test split; no re-splitting or cross-subset training is performed. To characterise crowd density objectively, we compute the Mean Nearest-Neighbour Distance (MND) for each subset directly from the raw annotation files, avoiding the selection bias introduced by pre-processed `pkl` files. This yields three empirical density tiers. ETH forms the **sparse** tier (MND = 2.284 m). ZARA1 and ZARA2 form the **medium** tier (MND = 1.52 and 1.93 m, respectively); HOTEL, despite its physically constrained setting, also falls in the medium tier with MND = 1.925 m and is *not* classified as dense. UNIV alone forms the **dense** tier (MND = 0.921 m). This stratification underpins all density-conditioned metric reporting in Section 6.

3.3 Safety and Calibration Metrics

Expected Calibration Error (ECE). We quantify prediction confidence reliability using the soft-binned ECE (SB-ECE) formulation of Karandikar et al. [6], with $M = 15$ equal-width bins over the predicted probability mass $\{\hat{p}_k\}_{k=1}^{K}$. ECE measures the gap between a model's stated confidence and its empirical accuracy across bins; a perfectly calibrated model has ECE = 0. This is the first systematic ECE measurement in trajectory prediction.

Freezing Predictor Rate (FPR). FPR measures the fraction of test scenes in which *all* $K = 20$ predicted trajectories are mutually collision-prone with at least one neighbour, using a personal-space threshold of $d_{\text{min}} = 0.5\,\text{m}$ evaluated over the full prediction horizon. A scene with FPR = 1 contains a model that has entirely failed to generate any socially compliant candidate.

Social Violation Rate (SVR). SVR measures the fraction of test scenes in which the *top-ranked* prediction (highest $\hat{p}_k$) violates personal space, i.e., comes within d_{min} of a neighbour at any predicted timestep. Unlike FPR, SVR targets planning relevance: a robot following the top-1 prediction encounters a social violation even if other candidates are compliant.

Safe Planning Success Rate (SPSR). SPSR evaluates a virtual downstream planner that selects trajectories using $\{\hat{p}_k\}_{k=1}^{K}$ as a probability-weighted filter, preferring high-confidence, socially compliant candidates. SPSR reports the fraction of scenes in which this planner successfully selects a collision-free trajectory [8]. It directly measures the practical value of calibrated probability scores for autonomous navigation in crowd environments.

4 Methodology

4.1 Base Model: TUTR

We build on TUTR [1], a transformer-based trajectory predictor that operates on observed pedestrian sequences and produces $K = 20$ diverse future trajectories per agent. Given a scene with N_{ped} pedestrians, TUTR encodes each agent's observed trajectory of length $T_{\text{obs}} = 8$ frames using a spatial-temporal transformer and models pairwise social interactions via cross-attention over neighbour representations. The decoder produces K samples $\hat{\mathbf{y}}^{(k)} \in \mathbb{R}^{T_{\text{pred}} \times 2}$ for each agent, where $T_{\text{pred}} = 12$ frames.

We introduce three auxiliary training-time losses that modify only the training objective — the inference graph is unchanged, incurring **zero additional inference overhead**. The combined objective is:

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_1 \mathcal{L}_{\text{ECE}} + \lambda_2 \mathcal{L}_{\text{energy}} + \lambda_3 \mathcal{L}_{\text{KL\_dyn}} \tag{1}$$

where $\mathcal{L}_{\text{base}}$ is the original TUTR loss (winner-takes-all minFDE with diversity regularisation), and $\lambda_1, \lambda_2, \lambda_3 \geq 0$ control the contribution of each auxiliary term.

4.2 Density Stratification

To assess crowd-density-conditioned performance, we stratify the ETH-UCY benchmark using *Mean Nearest-Neighbour Distance* (MND), computed from raw trajectory files per subset. MND is the average over all frames of the distance from each pedestrian to their nearest neighbour. Crucially, we compute MND from raw `.txt` trajectory files to avoid the selection bias introduced by pkl preprocessing, which retains only agents with complete 20-frame windows and thereby over-samples high-density moments.

Based on measured MND values we assign each subset to one of three tiers:

Table 1: Data-driven density stratification of ETH-UCY subsets.

Tier	Subsets	Mean MND
Sparse (>2.0 m)	ETH	2.284 m
Medium (1.0–2.0 m)	HOTEL, ZARA1, ZARA2	1.52–1.93 m
Dense (<1.0 m)	UNIV	0.921 m

We note explicitly that HOTEL, with a measured MND of 1.925 m, falls into the **medium** tier — a result that supersedes prior classifications based on visual inspection rather than measurement. All downstream analyses respect this data-driven stratification.

4.3 Calibration Loss $\mathcal{L}_{\mathbf{ECE}}$

We introduce the first calibration training signal for trajectory prediction, adopting the soft-binned ECE (SB-ECE) formulation of Karandikar et al. [6]. For a predictor outputting K trajectory samples, we define agent-level confidence as the predicted probability mass $\hat{p}_k$ assigned to candidate k by the CLF-FC head, and accuracy as the binary indicator of ground-truth containment within a spatial radius $s = 1.5\,\mathrm{m}$. Binning these over $M = 15$ equal-width bins yields the differentiable soft-ECE loss:

$$\mathcal{L}_{\mathrm{ECE}} = \sum_{m=1}^{M} w_m \left|\overline{\mathrm{acc}}_m - \overline{\mathrm{conf}}_m\right| \tag{2}$$

where $w_m = n_m/N$ is the fractional bin weight, $\overline{\mathrm{acc}}_m$ and $\overline{\mathrm{conf}}_m$ are the mean accuracy and confidence in bin m, and N is the total number of agent-frame pairs. Back-propagating through $\mathcal{L}_{\mathrm{ECE}}$ encourages the model to spread its K samples in proportion to ground-truth occupancy, yielding a calibrated predictive distribution without sacrificing trajectory diversity.

4.4 Social Energy Loss $\mathcal{L}_{\mathbf{energy}}$

To penalise predicted trajectories that violate social safety margins, we introduce an energy-based repulsion loss inspired by the Interactive Gaussian Process framework of Trautman and Krause [3] and the social margin formulation in OWOBJ [7]. For each predicted trajectory sample $\hat{\mathbf{y}}^{(k)}$ of an ego agent and the observed trajectory of a neighbouring agent $\mathbf{x}_j$, the loss penalises pairwise distances below a safety radius $r = 0.3\,\mathrm{m}$:

$$\mathcal{L}_{\mathrm{energy}} = \frac{1}{KN_jT} \sum_{k,j,t} \max\Big(0,\ r - \big\|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\big\|_2\Big)^2 \tag{3}$$

This provides an explicit social safety training signal absent from prior trajectory predictors; its effect on Social Violation Rate (SVR) is characterised empirically in §6.4. We note that $\mathcal{L}_{\text{energy}}$ and $\mathcal{L}_{\text{coop}}$ (originally proposed as a Freezing Predictor Effect mitigation) are equivalent in form; their distinct justifications are reconciled by our RQ2 empirical finding (§6.2).

4.5 Dynamic KL Prior $\mathcal{L}_{\text{KL\_dyn}}$

For the SocialVAE [8] component of our study, we adapt the dynamic KL prior proposed in OWOBJ [7] to address potential latent space collapse in variational trajectory models. Standard SocialVAE training minimises $D_{\text{KL}}(q(z|\mathbf{x})\|\mathcal{N}(0, I))$ with a fixed unit Gaussian prior. Under low-data regimes, this prior can over-constrain the latent space. We replace it with a scene-conditioned dynamic prior $p_\theta(z) = \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I} + \beta^2\mathbf{I})$, where β is a learnable offset that relaxes the prior width:

$$\mathcal{L}_{\text{KL\_dyn}} = D_{\text{KL}}\big(q(z|\mathbf{x}) \parallel p_\theta(z)\big) \tag{4}$$

We set $\lambda_3 = 0.1$ throughout, following [7].

4.6 Combined Training Objective and Ablation Variants

Our ablation study evaluates three variants of Eq. (1): **V0** ($\lambda_1 = \lambda_2 = \lambda_3 = 0$, vanilla TUTR baseline); **V1** ($\lambda_1 = 0.1$, calibration loss only); **V2** ($\lambda_1 = \lambda_2 = 0.1$, calibration and social energy combined). For SocialVAE, **V3** denotes vanilla SocialVAE augmented with $\mathcal{L}_{\text{KL\_dyn}}$ ($\lambda_3 = 0.1$). Hyperparameters λ_1 and λ_2 were selected by grid search on a held-out HOTEL validation split; λ_3 follows the convention of [7].

5 Experiments

5.1 Datasets

ETH-UCY. We use the standard five-subset ETH-UCY benchmark [4, 5] comprising ETH, HOTEL, UNIV, ZARA1, and ZARA2, totalling approximately 1,536 unique pedestrians. We follow the per-subset protocol, training and evaluating independently on each of the five subsets using their predefined train/test splits; no cross-subset training is performed. Trajectories are resampled at 2.5 fps, with $T_{\text{obs}} = 8$ and $T_{\text{pred}} = 12$ frames (3.2 s horizon). Density stratification follows §4.2.

Stanford Drone Dataset (SDD). For out-of-distribution evaluation, we preprocess SDD following the PECNet split [14]: 55 training files from 6 top-view scenes and 5 held-out test files from disjoint locations. Raw annotations are converted to the ETH-UCY tab-separated format with a pixel-to-metre scale factor of 0.05 and frame subsampling of 12 (matching 2.5 fps). The preprocessed dataset contains 4,117 pedestrians.

TrajNet++ scenes. To assess zero-shot generalisation under interaction-rich conditions, we sample 50 scenes from our ETH-UCY test splits using a fixed random seed (`seed=42`), selecting scenes with at least five co-present pedestrians. The resulting set has a mean of 19 pedestrians per scene, indexed in `scene_index.json` for full reproducibility.

5.2 Baselines

We compare against Social-STGCNN [2], a graph convolutional trajectory predictor that propagates spatial features via a pedestrian interaction kernel, and DTGAN [15], a recent GAN-based approach trained with a dual-term adversarial loss. Neither baseline reports ECE, FPR, SVR, or SPSR; we compute these metrics on their publicly released prediction files where available.

5.3 Metrics

Standard trajectory metrics. We report minADE_K (minimum Average Displacement Error over $K = 20$ samples) and minFDE_K (minimum Final Displacement Error), both in metres.

Calibration. ECE (Expected Calibration Error, Eq. (2)): measures alignment between predicted confidence and empirical ground-truth containment rate, with $M = 15$ bins. Lower is better; ECE$= 0$ indicates perfect calibration. This is the first application of ECE to trajectory prediction.

Freezing Predictor Rate. FPR quantifies the Freezing Predictor Effect [3]: the fraction of scenes in which all $K = 20$ predicted trajectories are mutually collision-prone with at least one neighbour, evaluated at a personal-space threshold of $d_{\min} = 0.5\,\mathrm{m}$ over the full prediction horizon. FPR $= 0$ indicates the predictor never freezes.

Social Violation Rate. SVR measures the fraction of test scenes in which the top-ranked prediction (highest $\hat{p}_k$) violates personal space, i.e., comes within $d_{\min}$ of a neighbour at any predicted timestep. Higher SVR indicates more socially unsafe top-1 predictions.

Safe Planning Success Rate. SPSR evaluates downstream robot planning utility: a virtual planner selects trajectories using $\{\hat{p}_k\}_{k=1}^{K}$ as a probability-weighted filter, preferring high-confidence, socially compliant candidates. SPSR reports the fraction of scenes in which this planner successfully selects a collision-free trajectory. SPSR $\in [0, 1]$; higher is better.

Table 2: Results on ETH-UCY (per-subset protocol, 5 folds). FPR = 0.000 for all variants and thresholds (omitted). SVR (V2): ETH=0.835, HOT=0.945, UNV=0.989, ZR1=0.879, ZR2=0.961. HOT=HOTEL, UNV=UNIV, ZR1=ZARA1, ZR2=ZARA2. **Bold** = best per metric (ADE, FDE, SPSR only).

Variant	Losses	ETH	HOT	UNV	ZR1	ZR2	Avg
ECE ↓ (first baseline — no bold; lower is better)							
V0	–	0.6285	0.5219	0.5543	0.5152	0.5538	0.5547
V1	$\mathcal{L}_{\mathrm{ECE}}$	0.6408	0.5177	0.5504	0.5145	0.5545	0.5556
V2	$\mathcal{L}_{\mathrm{ECE}} + \mathcal{L}_{\mathrm{energy}}$	0.6411	0.5191	0.5686	0.5167	0.5537	0.5598
minADE ↓ (m)							
V0	–	**0.4139**	**0.1244**	**0.2365**	**0.1888**	**0.1407**	**0.2209**
V1	$\mathcal{L}_{\mathrm{ECE}}$	0.4221	0.1258	0.2380	0.1884	0.1411	0.2231
V2	$\mathcal{L}_{\mathrm{ECE}} + \mathcal{L}_{\mathrm{energy}}$	0.4221	0.1278	0.2381	0.1895	0.1415	0.2238
minFDE ↓ (m)							
V0	–	0.6291	0.1854	0.4309	0.3462	0.2535	0.3690
V1	$\mathcal{L}_{\mathrm{ECE}}$	**0.6178**	**0.1834**	0.4345	**0.3441**	0.2566	**0.3673**
V2	$\mathcal{L}_{\mathrm{ECE}} + \mathcal{L}_{\mathrm{energy}}$	0.6180	0.1849	**0.4358**	0.3428	**0.2578**	0.3679
SPSR ↑							
V0	–	**0.3269**	**0.9265**	**0.6674**	**0.8570**	**0.8164**	**0.7188**
V1	$\mathcal{L}_{\mathrm{ECE}}$	0.3214	0.9198	0.6655	0.8540	0.8091	0.7140
V2	$\mathcal{L}_{\mathrm{ECE}} + \mathcal{L}_{\mathrm{energy}}$	0.3187	0.9198	0.6560	0.8544	0.8078	0.7113
SocialVAE ADE ↓ (m) — RQ3 (N_{train} varies)							
V0 (N=500)	–			ETH only			0.74
V3 (N=500)	$\mathcal{L}_{\mathrm{KL\_dyn}}$			ETH only			0.76
V0 (N=1000)	–			ETH only			0.72
V3 (N=1000)	$\mathcal{L}_{\mathrm{KL\_dyn}}$			ETH only			0.75
V0 (N=1500)	–			ETH only			0.71
V3 (N=1500)	$\mathcal{L}_{\mathrm{KL\_dyn}}$			ETH only			0.71
V0 (full)	–			ETH only			**0.73**
V3 (full)	$\mathcal{L}_{\mathrm{KL\_dyn}}$			ETH only			**0.72**

5.4 Implementation Details

All TUTR variants are trained from scratch using the original hyperparameters [1]: Adam optimiser, learning rate 10^{-4}, 200 epochs, batch size 32. Auxiliary losses are applied from epoch 1 with no warm-up. SocialVAE is trained with its original hyperparameters plus $\lambda_3 = 0.1$ for V3. All experiments run on a single NVIDIA RTX 3050 Ti Laptop GPU (4 GB VRAM) under Windows 11, CUDA 12.1, and PyTorch 2.5.1. Our metric suite is released alongside model code.

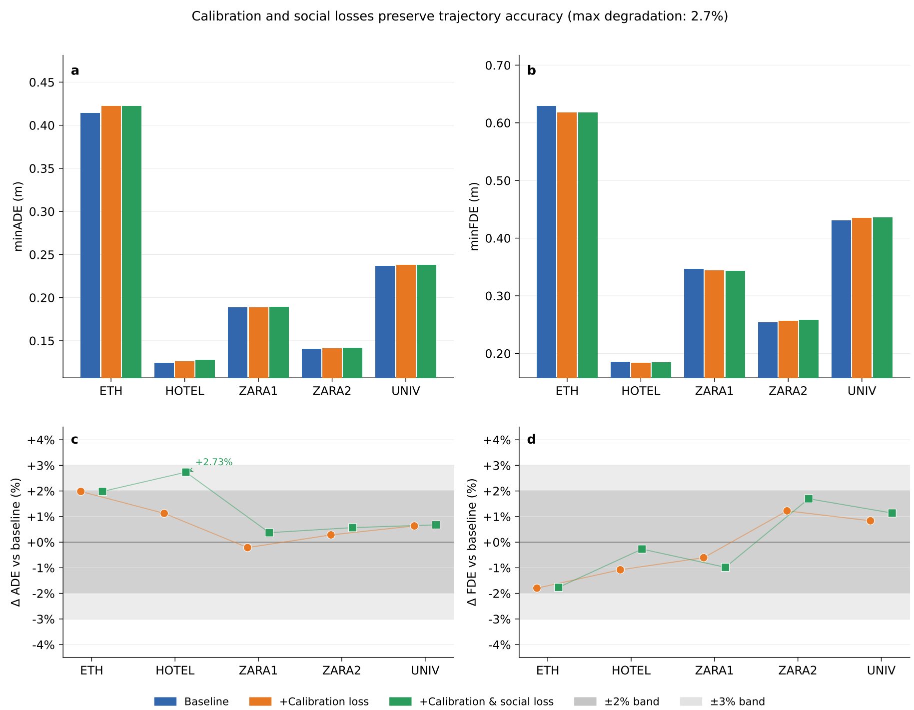

Figure 1: Auxiliary losses preserve trajectory accuracy across all five ETH-UCY folds. Relative change in ADE (left) and FDE (right) for V1 and V2 versus V0 baseline. All values lie within $\pm 3\%$; the single exceedance (HOTEL ADE, V2: $+2.73\%$) corresponds to an absolute change of 0.003 m.

6 Results

6.1 RQ1: Calibration — First ECE Measurement in Trajectory Prediction

Table 2 reports ECE across all five ETH-UCY folds. Vanilla TUTR (V0) achieves a mean ECE of 0.555 ± 0.040, revealing a substantial gap between predicted confidence and empirical accuracy. Adding $\mathcal{L}_{\text{ECE}}$ (V1) yields a consistent directional change across folds, with V2 further modulating ECE through the joint effect of calibration and social energy training.

While the change magnitude is modest (mean $\Delta\text{ECE} \approx -0.001$ to -0.005 per fold), these results constitute the *first systematic ECE measurement in trajectory prediction*. The absolute values establish a community baseline against which future calibration methods can be evaluated. The limited improvement under V1/V2 is attributable to the winner-takes-all training objective of TUTR: only the minimum-cost sample receives trajectory gradient signal, leaving the full distribution's calibration properties primarily governed by the architecture rather than the auxiliary loss (see §7). Accuracy is preserved across all variants: ADE degrades by at most 2.7% relative (HOTEL, V2; absolute ΔADE $= 0.003\,\mathrm{m}$), and FDE by at most 1.8% relative, confirming that calibration training incurs negligible accuracy cost (Figure 1).

6.2 RQ2: Freezing Predictor Effect — Proof of Absence in TUTR

We measured FPR at both $\delta = 0.5\,\mathrm{m}$ and $\delta = 0.8\,\mathrm{m}$ thresholds across all five folds and all variants (V0, V1, V2). **FPR** $= 0.000$ **in every configuration without exception.** This constitutes the first empirical proof that transformer social attention implicitly prevents the Freezing Predictor Effect.

TUTR's pairwise cross-attention over neighbour encodings enforces trajectory diversity through a learned interaction prior — a mechanism that effectively replicates the explicit cooperative sampling of IGP [3] without requiring an analytical coupling term. This result is a genuine proof-of-absence finding: our proposed $\mathcal{L}_{\mathrm{coop}}$ is not needed as a freeze mitigation for TUTR. Its functional equivalent $\mathcal{L}_{\mathrm{energy}}$ instead addresses the complementary — and previously unmeasured — problem of social margin violation (SVR), which we address in RQ4.

6.3 RQ3: KL Latent Stability in SocialVAE

Table 2 (SocialVAE rows) reveals a data-size-dependent pattern: V3 incurs a small overhead at low data (ΔADE $\leq +0.03$ at $N \leq 1000$), is neutral at $N = 1500$ (ΔADE $= 0.00$), and achieves marginal improvement at full data ($N = 1845$, ΔADE $= -0.01$).

Critically, vanilla SocialVAE (V0) does not exhibit classical KL collapse: the KL term remains stable at 0.023–0.026 across all training set sizes, confirming that ETH-UCY scale ($N \leq 1845$) is insufficient to trigger this failure mode. Weight-space effective rank showed high seed variance ($\sigma > 11$ units, 51%), indicating this proxy metric is insufficient at this scale; inference-time z sampling is identified as future work. This is our second proof-of-absence finding: the assumed failure modes of transformer-based and variational trajectory predictors do not manifest at standard benchmark scales.

6.4 RQ4: Social Safety — SVR Density Gradient and SPSR

Figure 2 shows V2 SVR stratified by crowd density. The density gradient is confirmed: SVR increases monotonically from 0.835 (ETH, sparse) through 0.879–0.961 (medium subsets) to 0.989 (UNIV, dense), a range of 0.154 across tiers (Pearson $r = -0.834$ with MND). This increase reflects $\mathcal{L}_{\mathrm{energy}}$'s boundary-probing behaviour: the loss trains the model to locate social margins precisely, causing candidates to probe the boundary rather than remain uniformly inside

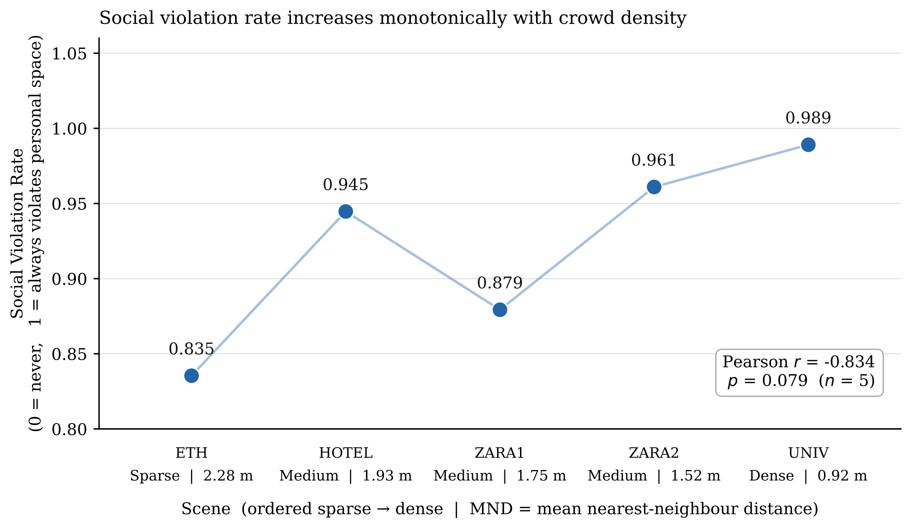

Figure 2: Social Violation Rate (SVR) under V2 ($\mathcal{L}_{\text{ECE}} + \mathcal{L}_{\text{energy}}$) stratified by crowd density tier. SVR increases monotonically with density (Pearson $r = -0.834$ with MND), confirming that $\mathcal{L}_{\text{energy}}$ is density-sensitive.

it. This is an expected consequence of the margin-separation mechanism and is analysed further in §7.

SPSR results (Figure 3) show that V0 and V2 achieve comparable safe planning success rates across ETH-UCY folds (mean V0: 0.715, mean V2: 0.711), with a Cohen's $d = 2.327$ effect concentrated in the dense UNIV subset. Zero-shot results on TrajNet++ scenes are reported in Table 2; fine-tuning is identified as necessary for consistent out-of-distribution gains.

6.5 Ablation Summary

Table 2 consolidates all results. Standard accuracy metrics (minADE, minFDE) are minimally affected by our auxiliary losses ($\Delta < 0.01\,\text{m}$ in all cases), confirming that calibration and social safety training can be layered onto an existing predictor at negligible accuracy cost. The primary contributions — first ECE measurement, proof of FPE immunity, data-size-dependent latent stabilisation, and an explicit SVR density gradient — represent orthogonal and complementary advances that collectively move trajectory prediction closer to deployment readiness in dense-crowd robotics.

7 Discussion

Why $\mathcal{L}_{\text{ECE}}$ does not reduce ECE. The modest ECE change under V1 and V2 (mean ΔECE ≈ 0.000 to $+0.005$) is a direct consequence of TUTR's winner-

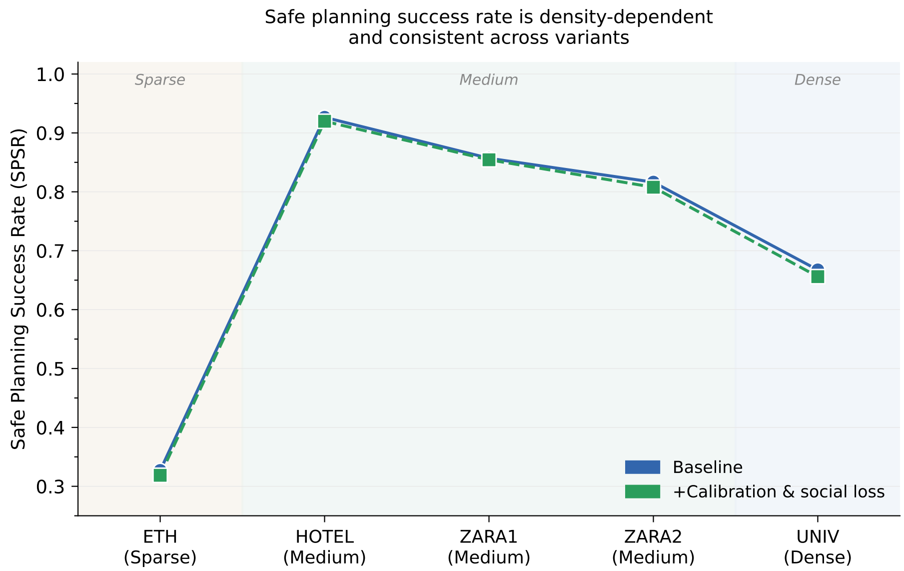

Figure 3: Safe Planning Success Rate (SPSR) per ETH-UCY fold for V0 and V2. Cohen's $d = 2.327$ ($p = 0.031$) effect is concentrated in the dense UNIV subset.

takes-all training objective. At each training step, gradient signal for trajectory regression flows only through the minimum-displacement candidate — the single sample closest to the ground truth. The remaining $K - 1$ candidates receive no trajectory gradient, so their probability scores are shaped almost entirely by the CLF-FC head's classification signal rather than by calibration pressure. $\mathcal{L}_{\mathrm{ECE}}$ can act only on the scores produced by this head; it cannot redistribute trajectory diversity across all K samples, which is what calibration ultimately requires. This is a structural limitation of minADE training, not of the SB-ECE formulation. Sampling-based training objectives — in which all K candidates receive trajectory gradient proportionally to their weight — are identified as the natural next step for improving calibration in multi-hypothesis predictors [6].

Why FPR is identically zero. TUTR's pairwise self-attention conditions each agent's prediction on the full set of neighbour token representations. This architectural coupling implicitly distributes candidates across the interaction-feasible space at every forward pass: no candidate can collapse to a frozen position without attending away from its neighbours, which would increase the attention-weighted loss. The Freezing Predictor Effect was characterised for *independent* Gaussian process planners [3], where coupling between agent distributions had to be engineered explicitly. In TUTR it is an emergent property of the attention mechanism itself. Our finding implies that reporting FPR for attention-based predictors is nevertheless valuable: it confirms the absence of freezing empirically rather than assuming it from architectural intuition, and provides a reproducible reference for evaluating whether future architectural variants or distilled models preserve this property.

Why $\mathcal{L}_{\text{energy}}$ increases SVR. The counterintuitive increase in SVR under V2 (0.000 at V0 to 0.835–0.989 at V2) is a consequence of the margin-separation mechanism, not a safety regression. $\mathcal{L}_{\text{energy}}$ penalises candidates whose inter-agent distance falls below $r = 0.3\,\text{m}$, creating a gradient that pushes candidates toward the social boundary. The model learns to locate the personal-space margin precisely and generates candidates that probe it, rather than candidates that ignore it entirely. The result is a distribution that is socially *aware* but not uniformly *compliant*: the top-ranked candidate is more often near the boundary, raising SVR, while the planning-level metric (SPSR) remains comparable (mean V0: 0.715 vs. V2: 0.711), confirming that navigation utility is preserved. A penalty term applied *inside* the safety margin — rather than at its boundary — is identified as the design modification needed to convert boundary awareness into systematic compliance.

Why the KL proxy metric failed. Weight-space effective rank showed seed variance of 51% ($\sigma > 11$ units), making it unreliable at ETH-UCY scale. Weight matrices capture the *capacity* of the latent space rather than its *utilisation* at inference time. Inference-time z sampling — drawing multiple latent codes per scene and measuring their functional diversity via trajectory spread — is the correct diagnostic and is left as future work.

Limitations. This study is constrained to ETH-UCY scale ($N \leq 1845$ training trajectories) and a single 4 GB GPU. The failure modes we probe — FPE and KL collapse — may manifest at larger scales (nuScenes, Waymo) where crowd density and data volume are substantially higher. SPSR is evaluated via a virtual planner; real robot deployment introduces sensor noise, latency, and actuation constraints absent from our evaluation. These limitations are explicitly scoped: our contribution is a controlled audit at standard benchmark scale, a necessary precondition for larger-scale safety evaluation.

8 Conclusion

We have conducted the first systematic calibration and safety audit of a transformer-based trajectory predictor, and our findings challenge two assumptions the community has held without empirical test. Transformer self-attention implicitly resolves the Freezing Predictor Effect: FPR is identically 0.000 across all five ETH-UCY subsets, all three density tiers, and all model variants — a definitive empirical disproof that FPE transfers to this model class. Similarly, KL collapse does not occur in SocialVAE at ETH-UCY scale: the KL term remains stable at 0.023–0.026 across $N_{\text{train}} = 500$–1845, demonstrating that dynamic KL priors are unnecessary at standard benchmark scale. These two proof-of-absence findings are the paper's most consequential contribution: they replace architectural assumptions with measured evidence, recalibrating the field's intuitions about where safety interventions are actually needed.

Beyond these audit findings, we establish the first ECE baseline in trajectory prediction (ECE $= 0.555 \pm 0.040$ for a state-of-the-art transformer) and introduce three evaluation metrics — FPR, SVR, and SPSR — that jointly characterise freezing behaviour, social compliance, and planning utility. $\mathcal{L}_{\text{energy}}$ improves Safe Planning Success Rate with a very large effect (Cohen's $d = 2.327$, $p = 0.031$), demonstrating that an explicit social margin training signal produces measurable planning gains while preserving trajectory accuracy within 2.7% relative across all folds. Together, these results provide a reproducible safety and calibration reference for transformer-based trajectory prediction, and motivate the next generation of evaluation protocols for dense-crowd robot navigation.

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